Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/19/2013
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	Maturing Technology for Major Reductions in Energy, Mass, and Crew Time for Plant Lighting in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	06/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date shows as 6/30/2014 per NSSC information (Ed., 5/6/14) NOTE: End date is now 7/31/2014 per NSSC information (Ed., 7/15/13) NOTE: new end date is 7/31/2013 per NSSC, and also is now showing start date of 6/9/2009 (Ed., 6/11/12) NOTE: new end date is 7/31/2012 per NSSC (Ed. 5/3/2011) NOTE: change in Period of Perfomance to 8/1/2009-7/31/2011 (from 6/9/2009-7/8/2011), per NSSC information (Ed., 3/11/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To develop and demonstrate proof of concept for two novel technologies that together will greatly advance the potential for growing food crops in space for crew life support at distant space destinations. The first technology is a “smart” LED-based sole-source lighting system for saving considerable energy for many reasons. The second technology is a crop-stand gas-exchange system consisting of a large cuvette, an open gas-exchange system, and computer controls for CO2, temperature, and atmosphere flow rate. Lighting parameters are controlled by a second computer. Using the smart lighting system in conjunction with the crop gas-exchange system will give real-time crop response of hydroponically grown crops to light (PPF, R/B ratio), temperature, and CO2 at all stages of crop production, which will define additional opportunities for significant energy savings during controlled-environment crop production.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project has broad impact for life on Earth as well as in space. On Earth, controlled environment agriculture (CEA) has limitations of profitability, economic viability, and affordability parallel to the issues that hold back the feasibility of using plants for crew life support in space. Interestingly, the single factor that limits the widespread use of CEA on Earth also is that of crop lighting! Supplemental and sole-source lighting using traditional high-intensity discharge (HID) lamps requires major electrical power consumption for waste-heat rejection as well as for lighting per se. The replacement of hot, inefficient HID lamps with solid-state light-emitting diode (LED) technology will benefit CEA and plant research for many reasons: LEDs are small, require much less mass and volume than HIDs, do not require massive ballasts, are extremely long-lived, are relatively cool at photon-emitting surfaces because waste heat is rejected elsewhere, emission wavelengths can be selected to match plant pigments, and can be operated at much lower power levels than HIDs because they can be placed in close proximity to plant tissues. As well, the field of LED technology is progressing at an ever-increasing rate, and electric-to-light conversion efficiencies are increasing at a rapid rate for LEDs. There is great interest in the use of LEDs for CEA on Earth as well as for crew life support in space. In addition, this project addresses the development of “smart” LED lighting systems that automatically sense plant position and size, and subsequently energize LEDs only directly above or in line with plant tissues. When used in conjunction with and supplemental to solar light in greenhouses on Earth, smart LED lighting systems also will monitor the instantaneous contribution of solar light as it tracks across greenhouses daily, and will automatically adjust LED switching and/or dimming/brightening controls in response to the total light environment. The crop gas-exchange system that is part of this project will be used in conjunction with smart LEDs to determine day-to-day crop tolerances, “preferences,” and limits for intensity of light, quality of light, and interaction of light with other important environmental parameters such as CO2 and temperature at all stages of crop development. This capability will greatly accelerate the optimization of environmental conditions needed to produce crops rapidly, efficiently, and affordably, which is just as important for CEA on Earth as it is for closed-loop life support in space.
Task Progress:	Summary: The graduate student (L. Poulet) who received a Master’s degree in May, 2012 optimizing spectra and reducing energy expenditure for growing plants hydroponically in a growth chamber using LEDs returned to Purdue in September following a summer space internship with the International Space University in Florida to repeat and refine experiments conducted during the M.S. thesis program in preparation for submitting a manuscript for publication to a space science journal. The goal of those follow-up experiments was to 1) minimize energy used for lighting during the lag phase of lettuce growth, which is an inherently non-productive latent period of plant development during the sigmoid growth curve, and to 2) enhance and optimize inputs during the accelerating exponential phase of growth, which can be highly productive, if optimized. Plants were grown for as long as the exponential phase lasts, but not beyond the inflection point into the plateau phase of the sigmoid growth curve, during which time productivity again decreases. A significant change from the originally intended objective of the approved FSB project was to switch detection of plant position and size from automatic HELIAC detection to manual, observational estimation of plant size and position. Ms. Poulet’s preliminary research indicated too many false positive signals caused by edge effects and structural cross bars on the array for the present version of HELIAC to be accurate enough for good detection of small plants, especially near the edges of arrays. Selective LED switching subsequently was done manually for the duration of the project. She summarized her observations and made recommendations to ORBITEC in an appendix of the thesis for subsequent versions of HELIAC with improved automatic detection, especially of small plants. Ms. Poulet completed four additional lettuce-cropping experiments between September and December of 2012 before departing for a staff position and to begin a PhD program in space life support in January, 2013 with DLR, the German Space Agency. Several versions of a manuscript detailing the energy-saving features of optimized, targeted lighting have been drafted and edited since Lucie’s departure from Purdue, and in June, 2013, a manuscript related to the most important aspects of her Master’s research was submitted to the journal Advances in Space Research. The manuscript has been peer reviewed and is in revision at the time of this reporting. At the end of December, 2012, design-build engineer Jerry Shepard delivered a custom-developed gas-exchange cuvette that clips over a hydroponics unit and around a HELIAC LED array. The cuvette system, which encloses a small crop stand growing hydroponically and is vertically expandable by means of bellows as the contained crop grows in height, was integrated with a computer-controlled gas-flow and CO2 injection/control system. The open gas-analysis system includes an absolute infrared gas analyzer (IRGA) for calibrating the mass-flow CO2 injection component against standard CO2 span gases as well as a differential IRGA to measure the difference in CO2 concentration between the return sample air stream from the outlet of the plant chamber (cuvette) and an airstream that bypasses the plant chamber. That CO2 difference multiplied by the gas-flow rate through the cuvette represents the photosynthetic rate of the crop stand growing therein. Throughout the time course of crop development, daily measurements of continuous CO2 removal rate divided by the surface area of the crop stand reflect the specific photosynthetic rate (Pn = µmoles CO2/m2/s) of the crop stand at the particular time in crop development that Pn is being measured. Upon termination of an experiment, lettuce or other test crop is harvested and plants are analyzed for leaf area and fresh and dry weight of leaves, stems, and roots. Thus, on the day of crop harvest, Pn also can be expressed as a function of leaf area. From January to May, 2013, EET senior David Lotz and ME junior Cannon Cheng began assembling and troubleshooting the integrated gas-exchange/hydroponics/lighting system. Initial efforts focused on sealing leaks from the cuvette that prevented return gas from reaching the gas analyzers. The outcome of the initial troubleshooting effort was to remove the cuvette from being mounted around the HELIAC LED array and place HELIAC on top of the cuvette with a transparent window on the top of the cuvette to transmit LED light into the plant compartment below. Thus, as the bellows expand the height of the cuvette to accommodate the contained, growing crop and increase internal cuvette volume, HELIAC is raised as well, keeping the light bank a constant distance above the top of the crop stand. Additional

	air leaks have been identified leading to the need for further minor modifications in how the cuvette is sealed against the rest of the integrated cuvette/hydroponics/HELIAC assembly, which will be extremely important for being able to slow the flow rate of air flow through the cuvette to be able to detect differential CO2 uptake when plants are still quite small. Two air-mixing fans were mounted within the cuvette to promote active mixing of air between inlet and outlet ports and the intervening crop canopy. Thermocouple sensors also were mounted within the cuvette internal space to be able to monitor light and dark ambient temperatures. A pan & tilt camera was mounted in one corner of the cuvette chamber to be able to visually monitor crop growth and status in real time without opening the cuvette and disturbing its internal environment. Both the hydroponics as well as the gas-flow plumbing were simplified. Use of flexible, thick-walled plastic tubing eliminated the need for junctions and fittings otherwise needed for bends and angle changes with inflexible tubing, thus many fewer potential leak points. David Lotz graduated from Purdue with a bachelor's degree in EET in May, 2013 and departed in June after accepting an engineering position with the Lense Corporation in Massachusetts. Cannon Cheng, who is interning with Allison Transmissions in Indianapolis during the summer, has continued his trouble-shooting work on the gas-exchange system some weekends during this summer.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Poulet L, Massa GD, Morrow RC, Bourget CM, Mitchell CA. "Smart Lighting for Major Reductions in Power and Energy for Plant Growth in Space." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Oral presentation. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 76. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012